403 HTTP Status Code in ASP.NET Core Web API

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In this article, I will discuss **How to Resolve 403 Forbidden HTTP Error Status Code in ASP.NET Core Web API**. We will also discuss **How to Resolve 403 Forbidden** error and various ways to implement **Role** or **Resource Based Authorization** in ASP.NET Core Web API with Examples.

What is 403 HTTP Status Code?

The HTTP 403 status code indicates "Forbidden". This means that the server understand the request but refuses to authorize it. This typically means that the client does not have the required permissions to access the requested resource, even though it has provided valid authentication credentials. This means the server is saying, "I understand your request, but I won't let you access the resource because you don't have the permissions to access the requested resource".

- **Insufficient Permissions:** The user or client does not have the required permissions to access the resource. This might be due to role-based access control settings or specific security policies.
- **IP Blocking:** The server may be configured to deny access based on IP addresses or ranges. If the client's IP address is blocked, the server will return a 403 error.
- **Resource Restrictions:** The server might restrict the requested resource, such as a file or directory that is not accessible to certain users.
- **CORS Violation:** Some servers implement Cross-Origin Resource Sharing (CORS) policies, which can result in a 403 error if the request violates these policies.

How to Resolve 403 Error

When encountering a 403 Forbidden error, users can attempt several steps to resolve the issue, including:

- **Check Permissions:** Ensure the user or client has the necessary permissions to access the requested resource. This may involve adjusting role-based access control settings or specific security policies.
- **Review Server Configuration:** Check the server configuration to ensure that the client is not blocking the error.
- **Check IP Restrictions:** Verify if there are any IP based restrictions that could be blocking access.
- **Clearing Cookies:** Sometimes, cookies can lead to a cached 403 error. Clearing cookies and browser cache can sometimes resolve the issue.

Note: Contact the website or server administrator for information on why access is denied or to request access if it is necessary as unusual legitimates be able to access.

How to Return HTTP Status Code in ASP.NET Core Web API?

In ASP.NET Core Web API, we can use a 403 Forbidden HTTP status code in several ways, depending on the context of an application and the specific requirements. Let's see the ways:

Using the **StatusCode method** to Return 403 HTTP Status Code without Data

In this, a 403 HTTP status code is returned without any data. This approach is suitable when the client only needs to know the Forbidden error. This method provides a convenient way to return an HTTP status code along with an optional message. The StatusCode method accepts parameters like Endpoint Controller or Actions, Here, we return a 403 Forbidden response without data using the StatusCode method.

```
[HttpGet]
[Route("GetEmployeeDetailsWithoutAuthorization")]
[ProducesResponseType(StatusCodes.Status403Forbidden)]
[ProducesResponseType(StatusCodes.Status200OK)]
[ProducesResponseType(StatusCodes.Status500InternalServerError)]
[ProducesResponseType(StatusCodes.Status400BadRequest)]
[ProducesResponseType(StatusCodes.Status404NotFound)]
[ProducesResponseType(StatusCodes.Status401Unauthorized)]
[ProducesResponseType(StatusCodes.Status405MethodNotAllowed)]
[ProducesResponseType(StatusCodes.Status406NotAcceptable)]
[ProducesResponseType(StatusCodes.Status408RequestTimeout)]
[ProducesResponseType(StatusCodes.Status415UnsupportedMediaType)]
[ProducesResponseType(StatusCodes.Status429TooManyRequests)]
[ProducesResponseType(StatusCodes.Status503ServiceUnavailable)]
[ProducesResponseType(StatusCodes.Status504GatewayTimeout)]
[ProducesResponseType(StatusCodes.Status505HTTPVersionNotSupported)]
[ProducesResponseType(StatusCodes.Status507InsufficientStorage)]
[ProducesResponseType(StatusCodes.Status511NetworkAuthenticationRequired)]
[ProducesResponseType(StatusCodes.Status422UnprocessableEntity)]
[ProducesResponseType(StatusCodes.Status423Locked)]
[ProducesResponseType(StatusCodes.Status424FailedDependency)]
[ProducesResponseType(StatusCodes.Status426UpgradeRequiredRequiredRequired
```